Functional Specification Document

Wedding Microsite & Fractional Sq.Ft. Gift Registry Platform • Engineering Release v1.0

STATUS: APPROVED • RERA/RBI COMPLIANT

1 SYSTEM ARCHITECTURE & CORE CONCEPT OVERVIEW

An enterprise celebration platform combining cinematic video invitations, multi-event ceremony RSVPs, and a real estate crowdfunding repository enabling guests to gift micro square feet of an apartment.

Pillar 1: Event Hub & Video
Cinematic invitation streaming, personalized URLs per family, wedding day countdown, and granular schedule itinerary for pre/post-wedding ceremonies.

Pillar 2: Smart RSVP Engine
Guest headcount validation, dietary preferences, accommodation logistics, automated WhatsApp confirmations, and digital photo wall.

Pillar 3: Fractional Registry
Interactive 1,200 SFT apartment floor-plan grid. Micro-unit contributions (1, 5, 10 SFT) mapped into an RBI-regulated tripartite escrow account.

2 SYSTEM ACTORS, ROLES & PERMISSION MATRIX

Actor / Role	Core Functional Scope & Capabilities	Permissions & Access Control
Host Couple / Family Primary Admin	Configures ceremony timings, uploads cinematic video invite, defines 1,200 SFT property target and base rate (₹/SFT), monitors real-time RSVP list.	Access to Couple Dashboard, broadcast WhatsApp dispatch trigger, guest list CSV import/export.
Invited Guests End User	Views multimedia invitation, books RSVP with dietary and accommodation requirements, pledges SFT units via UPI/Card, posts blessings.	Authenticated via dynamic URL / OTP token; read-only access to event data, transaction receipt generation.
On-Ground Event Team Venue Operations	Manages QR check-in desks at banquet halls, projects live-updating apartment donor wall, synchronizes reception slideshow stream.	Venue Console view, live websocket projection terminal, no access to financial ledgers or banking data.
Escrow Bank & Platform Ops Financial Admin	Monitors KYC validations, RERA builder linkage, reconciles virtual UPI handles, approves bulk settlement transfer to property developer.	Full tripartite escrow ledger access, settlement audit logs, dispute and refund overrides.

3 MODULE 1: EVENT HUB & CEREMONY MANAGEMENT

3.1 Cinematic Video & Dynamic Routing

- **Adaptive Bitrate Video Player:** 1080p H.264/WebM invite video with fallback CDN streaming.
- **Personalized Smart Links:** Dynamic query routing (`domain.com/invite?id=GID1024`) injecting personalized salutations.
- **1-Click Calendar & Maps:** Syncs individual ceremony timings to Google/Apple Calendar; embedded Google Maps API routing.

3.2 Granular Ritual Schedule Tracker

- **Pre-Wedding Track:** Pellikuthuru (Bride Haldi), Pellikoduku (Groom Haldi), and Mehendi/Sangeet Gala with song embeds.
- **Muhurtham Track:** Mangalasnan, Kanyadaan, and Lagna Muhurtham dawn countdown with low-latency RTMP live-stream.
- **Reception Track:** Evening dinner timings, VIP seating guidance, and photo-booth locator.

4 MODULE 2: SMART RSVP, GUEST LOGISTICS & DIGITAL WALL

4.1 RSVP Intake & Logistics Subsystem

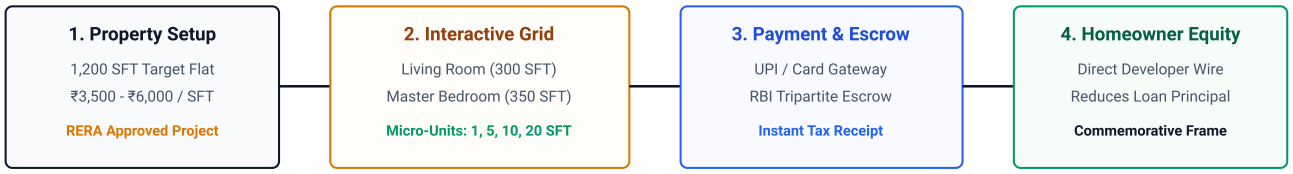
- **Headcount Breakdown:** Collects discrete counts for adults, children, and infants per registered party.
- **Dietary Segmentation:** Multi-select taxonomy: South Indian Traditional Veg, Jain, Vegan, Non-Veg, Nut/Gluten Allergies.
- **Transit & Accommodations:** Intake fields for arrival flight/train schedules, shuttle booking, and hotel check-in dates.
- **Automated WhatsApp Webhooks:** Immediate two-way WhatsApp notification confirming booking status and venue pin.

4.2 Digital Guestbook & Photo Stream

- **Wishes Engine:** Text blessing module with profanity filters, stored against guest ID and timestamp.
- **Direct Camera Upload:** Progressive web uploader allowing guests to snap and upload photos in real time.
- **Host Moderation Queue:** Admin toggle to review, approve, or hide images prior to public projection.
- **WebSocket Broadcast:** Low-latency socket emitting approved blessings to banquet hall projection screens.

5 MODULE 3: FRACTIONAL SQ.FT. REAL ESTATE REGISTRY ENGINE

The core architectural innovation converts a 1,200 SFT residential floor plan into an interactive crowdfunding canvas divided into purchaseable micro-units.



5.1 Floor Plan State Management & Data Schema

Tile State	Visual Representation	Database Trigger & Logic	User Action Allowed
AVAILABLE	White background, slate borders, hover gold accent.	Default state (`is_funded = FALSE`, `locked_until = NULL`).	Select denomination (1, 5, 10 SFT), add to gift cart.
RESERVED (IN-CART)	Amber hatched fill with animated countdown.	Atomic distributed lock set for 10 minutes (`locked_until = NOW() + 600s`).	Proceed to checkout or release back to public pool.
FUNDED / LOCKED	Solid gold leaf fill with donor initials and tooltip.	Escrow payment gateway callback success (`txn_status = 'SETTLED'`).	Read-only; click to view donor message and blessing tag.

6 MODULE 4: FINANCIAL COMPLIANCE, ESCROW & SETTLEMENT

6.1 Tripartite Escrow Pipeline

- **RBI Compliant Escrow:** All funds flow directly into an RBI-regulated commercial bank virtual escrow account, preventing commingling.
- **Direct RERA Builder Disbursement:** Settled funds are transferred via RTGS directly to the builder's verified RERA project escrow account or couple's home loan account.
- **Dual Authorization:** Release of funds requires cryptographic sign-off from both host primary and platform compliance officer.

6.2 Automated WhatsApp Receipts & Tax Framework

- **Wedding Gift Exemption:** Invoices generated noting tax-exempt gifts received on the occasion of marriage under Section 56(2)(x) of the Indian Income Tax Act.
- **Instant Dispatch:** PDF receipt with transaction hash sent to donor within 5 seconds of gateway success.
- **Automated Video Note:** Triggers pre-recorded personalized video thank-you note from couple to guest's WhatsApp.

7 MODULE 5: ON-GROUND WEDDING OPERATIONS & LIVE COUNTERS

Operational Unit	Hardware & Network Deployment	Functional Output
Banquet QR Stands	Matte acrylic stands with unique dynamic QR codes positioned at dining tables, gift counters, and entry arches.	1-second scan-to-registry routing; automatically pre-populates table number in guest blessing tag.
Live Stage LED Projection	1080p/4K HDMI output driven by dedicated venue laptop connected to local cellular 5G hotspot.	Dynamic apartment progress bar (e.g., 420/1,200 SFT funded), rotating guest photos, and real-time blessings ticker.
Founding Wall Memento	Post-event automated vector render sent to high-resolution physical printing studio.	Framed physical apartment blueprint listing every contributing guest engraved onto their funded square feet.

8 NON-FUNCTIONAL PERFORMANCE & SECURITY SPECIFICATIONS

Metric Category	Standard Specification Requirement	Validation Benchmark
Latency & Rendering	Initial page render < 1.8s; dynamic 1,200 SFT floor plan grid render < 800ms over 4G/5G mobile.	Lighthouse Performance Score ≥ 90; edge-cached assets via Cloudflare CDN.
Peak Concurrency	Support 5,000+ simultaneous guest connections during banquet hours without transaction lockouts.	Redis distributed lock with auto-expiry; horizontally auto-scaled container pods.
Data Security	End-to-end 256-bit TLS 1.3 encryption, PCI-DSS Level 1 compliant gateway, AES-256 for PII storage.	Zero plain-text financial storage; tokenized UPI/VPA and automated daily database snapshot.
Browser Compatibility	Responsive across iOS Safari (v14+), Android Chrome (v90+), Desktop Edge, Chrome, Safari, Firefox.	Zero horizontal scroll; responsive grid viewport adapting from 360px mobile to 4K LED screens.